

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 113: Discrete Mathematics

Batch: 2020

Semester: Fall 2025

Date: 03/01/2026

Mid-Term Examination

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions.

5 x 6 = 30

1. (a) Let $U = \{1, 2, 3, 4, 5, 6, 7\}$, $B = \{1, 2, 3, 4\}$ and $C = \{5, 6, 7\}$ be 3 sets. CLO 4 2
Determine $B \times C$ and prove that $B' = C$.
(b) Let U be a universal set that contains all alphabets in small letters. V be a set CLO 4 4
that contains all vowels and W be a set that contains all consonant between
first two vowels. Show the complement of V in Venn diagram. Find the
power set of V and cardinality of the complement of V . ✓
2. (a) Write the differences between predicate calculus and propositional calculus. CLO 1 2
(b) Let $f_1(x) = x^3 - 2x^2 + 1$ and $f_2(x) = x^3 + 4x - 2$. Determine the types of given CLO 4 4
functions with reason. Find $f_1 f_2$ and $f_1 + f_2$.
3. (a) Determine the general equation for the Tower of Hanoi Problem. CLO 4 3
(b) What is meant by Recurrence relation? Find the first five terms of the CLO 4 3
sequence defined by the recurrence relation and initial condition given
below: $a_n = na_{n-1} + n^2 a_{n-2}$, $a_0 = 1$, $a_1 = 1$.
4. (a) Write short notes on Exclusive OR and Tautology of a Proposition. CLO 1 2
(b) By using mathematical induction technique, prove that CLO 1 4
$$\sum_{j=0}^n ar^j = \frac{ar^{n+1} - a}{r - 1}$$
5. Find the truth values for the following expression: CLO 1 6
 $(((\neg p \vee q) \wedge (\neg p \vee r)) \rightarrow ((q \wedge r) \rightarrow (p \rightarrow \neg r))) \rightarrow \neg q$, and justify whether the
given expression is either a Tautology or not.
6. Determine the expressions $\{[p \wedge \neg(\neg p \vee (q \oplus \neg r)) \leftrightarrow (p \wedge q)] \rightarrow p\}$ and CLO 1 6
 $[\neg(p \vee (p \oplus \neg q)) \leftrightarrow (q \wedge \neg r)]$ are either logically equivalent or not.
7. Let $A = \{1, 2, 4, 12, 24, 48\}$ be a set, find the following relations: CLO 4 6
 $R_1 = \{(a, b) \mid a \leq b\}$,
 $R_2 = \{(a, b) \mid a \text{ divides } b\}$ and
 $R_3 = \{(a, b) \mid a^2 + b \leq 200\}$
Find the properties with reason of each relation. Represent each relation in matrix
form and graphically.

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 117: Engineering Ethics and Cyber Law

Batch: 29th

Semester: Fall 2025

Date: 05/01/2026

Midterm Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

1. (a) What is Moral Dilemma? CLO1 3
(b) Outline the differences between Morality and Ethics. CLO1 3

2. Alex, a system engineer, discovers that his company's new IoT device collects more personal user data than disclosed in their privacy policy. The data is being sold to third parties without user consent. When he raises this concern, his managers argue that all tech companies do this and it's essential for their business model.
(a) Evaluate this situation using the three types of inquiries. CLO2 3
(b) Under what conditions would whistleblowing be ethically justified in this case? Support your answer with ethical principles CLO2 3

3. Think about a scenario, you are giving exam on this course and you didn't study the chapters properly. So naturally you are biting your pen instead of writing by it on the paper. Now suddenly you see your that same friend who told you last night that "Mama, kichuuiiii Pori nai. Kal Sure fail" is writing smoothly at the speed of light!! You also notice that the secret of his success is not his study rather his ability to do copy and bring 'Nokol' like a pro in the exam hall. CLO2 6

Now explain this real-life part of yours from the 3 normative ethical approaches.

4. (a) Explain the concept of duty ethics with a practical example from engineering. CLO2 3
(b) Explain with an example how virtue ethics differs from deontology. CLO2 3
5. Explain the Lawrence Kohlberg's Theory on moral development. CLO2 6

A manager says engineers must be "accountable" for safety issues. One engineer asks, "Does accountability mean legal or moral responsibility?"

- (a) What type of inquiry is being used here? Explain with proper examples CLO2 3
(b) Why is this clarification important? Explain how to approach for that. CLO1 3

decision
decision
decision
decision
decision

~~7~~

- (a) Explain Aristotle's Golden Mean in details.
- * (b) How does MacIntyre define virtues in the context of professional practices?

CLO2 3

CLO2 3

$$\sin \Rightarrow \frac{2\pi}{30} \times 5 + \frac{\pi}{6}$$

Department of Electrical and Electronics Engineering
Faculty of Science, Engineering and Technology

Course Code: PHY 121
Mid-Term Examination
Date: 7/1/2026

Total Time: 1:30 Hours

Course Title: Physics I
Semester: Fall 2025
Total Marks: 30

Answer any three (03) out of the following questions:

3 × 10 = 30

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- | | | CLOs | Marks |
|---------|--|-----------|-------------|
| Q1. (a) | State Hook's Law. From this law derive the differential equation for Simple Harmonic Motion (SHM) and solve the equation to obtain an expression for the displacement of a particle executing SHM. | CLO1
+ | [6]
CLO2 |
| (b) | Show that the mean kinetic and potential energies of non-dissipative simple harmonic vibrating particle systems are equal. | CLO2 | [4] |
| Q2. (a) | For a two body oscillator system, prove the relation, $T = 2\pi\sqrt{\mu/k}$, where, $\mu = \frac{m_1 m_2}{m_1 + m_2}$ | CLO2 | [4] |
| (b) | A particle executes simple harmonic motion given by the equation- $y = 20 \sin(\omega t + \alpha)$. If the time period is 30 seconds and the particle has a displacement of 10 cm at $t = 0$ sec, find (i) epoch, (ii) the phase angle at $t = 5$ sec and (iii) the phase difference between two positions of the particle 15 seconds apart. | CLO3 | [6] |
| Q3. (a) | Define the following term-
i) Coherent sources
ii) Distortion | CLO1 | [2] |
| (b) | Show that for a simple pendulum, time period $T = 2\pi\sqrt{\frac{l}{g}}$ | CLO2 | [4] |
| (c) | A student at HUB is investigating simple harmonic motion in a Physics Lab. An oscillating mass on a spring is used in the investigation. The student measures that a body of mass 0.5 kg is suspended from a spring, stretching it by 0.07 m. For a displacement of 0.03 m, the body has a downward velocity of 0.4 m/s. Compute the following:
(i) the time period of the oscillation
(ii) the frequency of the oscillation
(iii) the amplitude of the vibration of the spring | CLO3 | [4] |

$$m = 0.5$$

$$x = 0.07$$

$$y = 0.03$$

$$v = 0.4$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$k = \frac{mg}{x} = 70 \text{ N m}^{-1}$$



Q4

- (a) Define Simple Harmonic Motion and write down the conditions for SHM. CLO1 [4]
- (b) The scale of spring of a spring balance reading from 0-10 kg is 0.25 m. A body suspended from the balance oscillates with a frequency of $\frac{10}{\pi}$ hertz. Calculate the mass of the body attached to the spring. CLO3 [3]
- (c) Find whether the discharge of a condenser through the following inductive circuit is oscillatory? Also find the resonant frequency. CLO3 [3]
Where, $C = 0.1 \mu\text{F}$, $L = 10 \text{ mH}$ and $R = 200 \Omega$

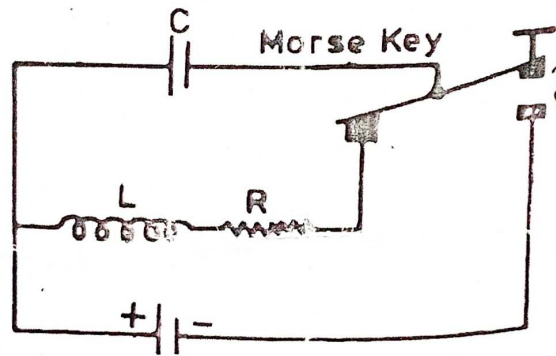


Fig: 5-1

- Q5. (a) Two equal masses m are connected with the two identical massless springs each of spring constant K . Show that the angular frequencies are given by- CLO2 [6]
- $$\omega_1^2 = (3 + \sqrt{5}) \frac{k}{2m}$$
- $$\omega_2^2 = (3 - \sqrt{5}) \frac{k}{2m}$$
- (b) In a biprism experiment with sodium light, bands of width 0.0195 cm are observed at 100 cm from the slit. On introducing a convex lens 30 cm away from the slit, two images of the slit are seen 0.7 cm apart, at 100 cm distance from the slit. Calculate the wave length of sodium light. CLO3 [4]



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering

Faculty of Science, Engineering and Technology

Course Code: MAT115

Course Title: Mathematics I (Differential and Integral Calculus)

Midterm Examination

Semester: Fall 2025

Date: 1/10/2026

Total Time: 1:30 Hours

Total Marks: 30

Answer any three (03) out of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No	Question	CLOS	Marks
Q1.	(a) Define function, domain and range.	CLO1	[3]
	(b) Find out Domain and Range of the following functions. (i) $y = \sqrt{x-4}$ (ii) $y = \ln \frac{x+1}{x-1}$	CLO3	[5]
	(c) Find the inverse function of $f(x) = \log_e(x + \sqrt{x^2 - 1})$.	CLO4	[2]
Q2.	(a) What do you mean by even, odd and periodic function.	CLO1	[3]
	(b) Draw the graph of the following functions. (i) $f(x) = \begin{cases} x^2 & \text{where } x < 0 \\ x & \text{where } 0 \leq x \leq 1 \\ \frac{1}{x} & \text{where } x > 1 \end{cases}$ (ii) $f(x) = \begin{cases} x & \text{where } x \neq 0 \\ 0 & \text{where } x = 0 \end{cases}$	CLO2	[3]
	(c) Check the limit exist or not. (i). $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$ (ii). $\lim_{x \rightarrow \infty} x - \sqrt{x^2 - x}$ (iii). $\lim_{x \rightarrow 0} \left(x - \frac{x}{5}\right)^{\frac{1}{x}}$	CLO3	[4]
Q3.	(a) What do you mean by Continuity and Differentiability?	CLO1	[2]
	(b) Show that $f(x) = \begin{cases} x^2 & \text{where } x < 0 \\ x & \text{where } 0 \leq x \leq 1 \\ \frac{1}{x} & \text{where } x > 1 \end{cases}$ is continuous at $x = 1$ and	CLO4	[4]
	(c) Check Differentiability of the following function $f(x) = x + 2 - x + x - 1 $.	CLO3	[4]
Q4.	(a) State and prove that Leibnitz's Theorem.	CLO2	[4]
	(b) If $\sin^{-1} y = k \sin^{-1} x$. Then show that $(1 - x^2)y_{n+2} - (2n + 1)y_{n+1} + (k^2 - n^2)y_n = 0$.	CLO3	[4]
	(c) Successive differentiate of the function $f(x) = \sin 2x \cos x$ in term of n .	CLO3	[2]
Q5.	(a) Define slope, Tangent line.	CLO1	[2]
	(b) Evaluate Slope, tangent line, normal line of the following function $y(x - 2)(x - 3) - x + 7 = 0$ at $(2, 1)$.	CLO3	[4]
	(c) Calculate the maximum and minimum values of the following function. $f(x) = x^3 - 6x^2 + 9x + 1$.	CLO3	[2]
	(d) If $x = a(\theta + \sin \theta)$ and $y = a(1 - \cos \theta)$ then prove that $\frac{d^2y}{dx^2} = \frac{1}{4a} \sec^4 \frac{\theta}{2}$.	CLO3	[2]

$$D = \mathbb{R} - \{1\}$$

$$R = \mathbb{R} - \{0\}$$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

ENG 111: English Basics

Mid-term Examination
Fall 2025

29th Batch

1st Semester

12 January 2026

Time: 01 Hour and 30 Minutes

Total Marks: 30

CLOs

Marks

PART A

CLO1 01. a) Write short notes on the following:

5 X 2= 10

- i. Scanning
- ii. Intensive Reading

Or

b) How can you use reading as a primary method for vocabulary enhancement?

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PART B

Read the passage below and answer the following questions:

Car companies have used robots to build cars since the 1970s. Now, a new challenge has arrived: automation inside the cars themselves. While self-driving technology has existed for 50 years, progress became much faster after Google began testing its self-driving cars in 2010. There are three main reasons for this technology: Safety, Free Time, and Independence. Over 90% of car accidents are caused by human error. Automation could stop these crashes. People can relax, work, or talk while the car drives itself. It helps elderly or disabled people travel more easily.

Right now, most cars sit parked 90% of the time. In the future, people might stop owning personal cars and use car-sharing services. This would mean we need much fewer vehicles on the road to get everyone where they need to go. Research shows mixed results for car usage. One study suggests we might only need 30% of the cars we use today. However, another study says that while fewer people will own cars, the cars we do have will be driven much more often. Because they are used so much, they will wear out faster and need to be replaced more frequently.

Instead of owning one car that does everything, people might use a "mobility provider." This allows them to pick a small, efficient car for a solo trip or a large vehicle for a family camping trip. There are still several challenges to overcome: making cars work in bad weather and heavy traffic, deciding who is responsible for an accident when there is no driver, getting people to trust and accept robot cars. Experts believe these problems can be solved in the next 10 years. While it is hard to predict exactly what will happen, the way we move is definitely going to change.

CLO2 02. Choose the most suitable option for each of the following questions.

1 X 5 = 05

a) Why is self-driving technology being developed?

- i. To make sure cars are replaced every year.
- ii. To help companies use more robots in factories.
- ☒ iii. To provide independence for travellers.
- iv. To stop people from going on camping trips.

b) What does 'mobility provider' mean in the text?

- i. A company that fixes broken tires on the road.
- ☒ ii. A service that lets you pick a specific car for a specific trip.
- iii. A person who helps the elderly walk to their cars.
- iv. A type of engine that makes cars go faster in traffic.

c) How does car usage in the future differ from today?

- i. Future cars will be owned by individuals.
- ☒ ii. Future cars may be shared and driven much more.
- iii. There is no difference.
- iv. In future, numerous cars will remain parked.

d) What does the 'family camping trip' example show?

- i. That self-driving cars cannot drive in the woods.
- ii. That mobility providers only offer small, efficient cars.
- iii. That families will stop owning tents in the future.
- ☒ iv. That people will need a large vehicle for specific trips.

e) Why is 'free time' mentioned as a reason for this technology?

- i. Because people will need more time to fix their cars.
- ii. Because it takes longer to get to work in a robot car.
- iii. Because people can talk while the car drives.
- iv. Because robot cars only drive during the holidays.

CLO2 03. Fill in the blanks with appropriate words.

1X 5= 05

- a) Automated cars are important because they can stop crashes caused by human *error*.
- b) Instead of owning individual cars, people might use a system of *car sharing* services.
- c) Experts think robot cars will be able to drive in bad *weather and heavy traffic* within 10 years.
- d) Computers can drive cars, but it is hard to get people to *trust* the robots.
- e) Because shared cars are driven so much, they will need to be *replaced* more often.

CLO2 04. Answer the following questions providing information from the passage.

2 X 5= 10

- a) Why is self-driving technology an important invention?
- b) How might people travel in the future instead of owning a car?
- c) Why can't robot cars drive everywhere easily right now?
- d) According to the passage, what is hard for people to give to robot cars?
- e) How do "mobility providers" help people pick the right vehicle?



HAMDARD UNIVERSITY BANGLADESH

Department of Electrical and Electronics Engineering
Faculty of Science, Engineering and Technology

Course Code: EEE117
Mid-Term Examination
Date: 1/14/2026

Course Title: Electrical Engineering
Semester: Fall 2025
Total Marks: 30

Total Time: 1:30 Hours

Answer any three (03) out of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- Q1. (a) Clasify the sources in electrical circuits with proper symbols.
(b) A resistor absorbs an instantaneous power of $20 \cos^2 t$ mW when connected to a voltage source $v = 10 \cos t$ V. Find i and R . $i = 2$, $R = 5 \cos t$.
(c) Obtain the equivalent resistance for the circuit in Fig. 1(c) and use it to find current i . 16.06

CLOs	Marks
CLO1	[3]
CLO3	[3]
CLO3	[4]

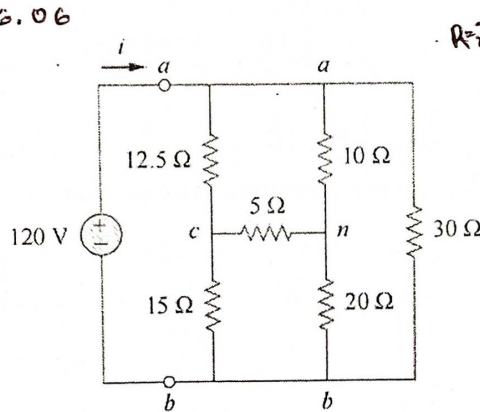


Fig. 1(c)

- Q2. (a) Write short note on: i. Supernode, ii. Supermesh
(b) Find the voltages at the three nonreference nodes in the circuit of Fig. 2(b).

CLO1	[2]
CLO3	[4]

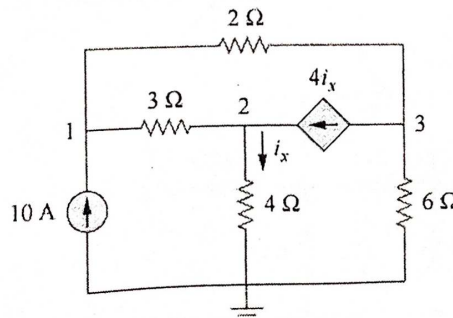


Fig. 2(b).

(c) Use mesh analysis to find the current I_0 in the circuit of Fig. 2(c).

CLO3 [4]

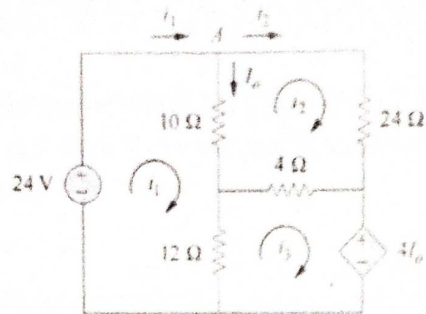


Fig. 2(c).

$I_0 = 2.04$

- Q3. (a) Write short on Superposition principle and source transformation.
(b) For the circuit in Fig. 3(b), use the superposition theorem to find i .

CLO1 [2]

CLO3 [4]

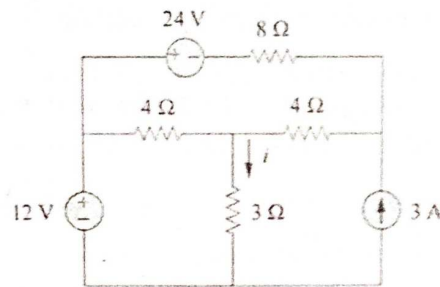


Fig. 3(b)

- (c) Find i_0 in the circuit of Fig. 3(c) using source transformation.

CLO3 [4]

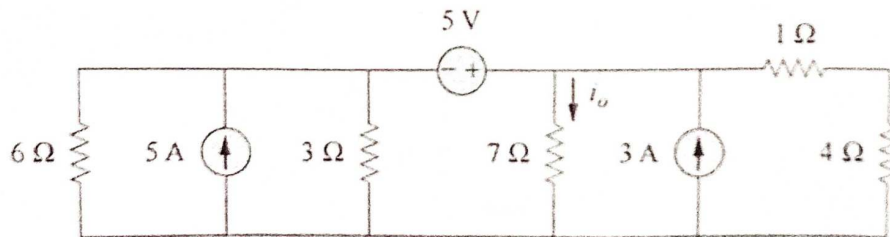


Fig. 3(c)

$i = \frac{V}{R}$
 $R \cdot V = IR$

- Q4. (a) State Thevenin and Norton's theorem.
(b) Find the Thevenin equivalent of the circuit in Fig. 4(b) at terminals $a-b$.

CLO1 [2]

CLO3 [4]

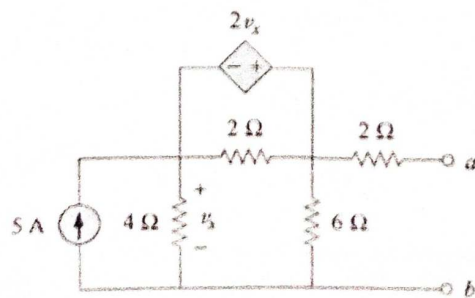


Fig. 4(b)

- (c) Find the Norton equivalent circuit for the circuit in Fig. 4(c), at terminals a - b .

CLO3 [4]

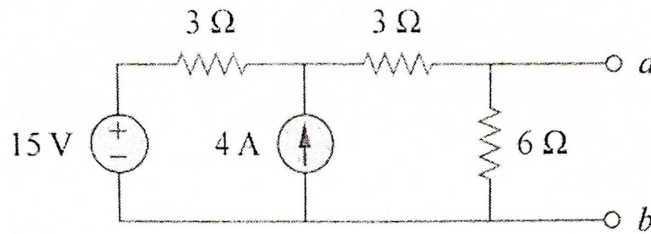


Fig. 4(c)

- Q5. (a) State Maximum Power Transfer Theorem CLO1 [2]
 (b) Show that the maximum power is transferred to the load when $R_L = R_{Th}$, and prove that under this condition the power transferred is $P_{max} = \frac{V_{Th}^2}{4R_{Th}}$. CLO2 [4]
 (c) Determine the value of the load resistance R_L required for maximum power transfer in the circuit shown in Fig. 5(c). Also, calculate the maximum power transferred to the load. CLO3 [4]

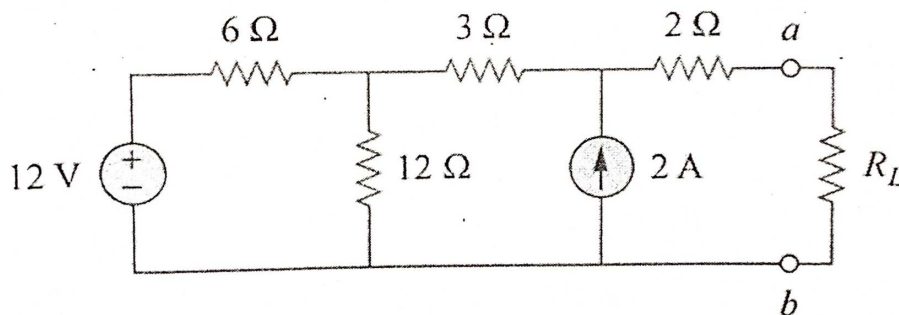


Fig. 5(c)